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DISPLAY PANEL AND MOBILE TERMINAL WITH AT LEAST ONE SUB-RECESS

Abstract

The present application discloses a display panel and a mobile terminal. The display panel includes a substrate, a light emitting layer, a touch layer, a first refractive index layer, and a second refractive index layer. A refractive index of the second refractive index layer is greater than a refractive index of the first refractive index layer. The touch layer comprises touch wirings extending from a display region to a non-display region. A first type recess in the first refractive index layer covers the touch wirings. The second type recess includes at least one sub-recess not overlapping the touch wirings.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/623,893 filed on Dec. 30, 2021, which is a National Phase of PCT Patent Application No. PCT/CN2021/140146 having International Filing Date of Dec. 21, 2021, which claims the benefit of priority of Chinese Patent Application No. 202111552189.1 filed on Dec. 17, 2021. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

FIELD OF INVENTION
[0002] The present application relates to a field of display technologies, especially to a display panel and a mobile terminal.
BACKGROUND OF INVENTION
[0003] At present, to lower a power consumption of an organic light emitting diode (OLED) display panel and improve an efficiency of the OLED display panel, display panel manufacturers constantly set forth new technologies. For example, by geometric optics, a micro lens pattern (MLP) is disposed in an OLED screen body to converge diffused light emitted from the OLED screen body to a place right above a screen body, which is one of effective solutions of improving a light emission rate of the OLED display panel.
[0004] However, the micro array pattern in the MLP technology usually needs to be manufactured by an inkjet printing process to make it flat for convenience of later processes. The inkjet printing process requires to set a slit around a periphery of the screen body in advance to block flow of an ink and restrict a coverage of the ink. Because touch wirings and a MLP film layer are disposed adjacent to each other and an overlapped exists between the touch wirings and the slit, over etching occurs when the slit is etched to such that the touch wirings are exposed to result in a technical issue of touch failure.
SUMMARY OF INVENTION
Technical Issue
[0005] The present application provides a display panel and a mobile terminal to solve a technical issue that touch failure easily occurs in touch wirings in a conventional display panel.
Technical Solution
[0006] The present application a display panel, the display panel comprises a display region and a non-display region located on at least one side of the display region; the display panel further comprises: [0007] a substrate; [0008] a light emitting layer disposed on a side of the substrate, wherein the light emitting layer comprises a plurality of light emitting pixels disposed in the display region; [0009] a touch layer disposed on a side of the light emitting layer away from the substrate, wherein the touch layer comprises touch electrodes disposed in the display region and touch wirings electrically connected to the touch electrodes, and the touch wirings extend from the display region to the non-display region; [0010] a first refractive index layer disposed on a side of the touch layer away from the substrate, wherein the first refractive index layer comprises a plurality of first apertures distributed in the display region and corresponding to the light emitting

pixels; [0011] a second refractive index layer disposed on a side of the first refractive index layer away from the substrate, filled in the first apertures, and a refractive index of the second refractive index layer is greater than a refractive index of the first refractive index layer; [0012] wherein the first refractive index layer further comprises a plurality of barrier notches distributed in the non-display region and extending along a first direction, the barrier notches comprise a first type recess and a second type recess, the first type recess is disposed to cover the touch wirings, and the second type recess comprises at least one sub-recess not overlapping the touch wirings.
[0013] In the display panel of the present application, the display panel further comprises: [0014] a driver circuit layer disposed between the substrate and the light emitting layer, wherein the driver circuit layer comprises a plurality of driver circuits electrically connected to corresponding ones of the light emitting pixels and driver wirings electrically connected to the driver circuits, and the driver wirings extend from the display region to the non-display region; [0015] wherein the non-display region comprises a bending region and a wiring exchange region disposed between the bending region and the display region, wherein the touch wirings are electrically connected to the driver wirings through via holes in the wiring exchange region; [0016] wherein a wiring exchange region is disposed between adjacent two of the sub-recesses.
[0017] In the display panel of the present application, an overlapped position between the first type recess and the touch wirings are filled with a covering portion.
[0018] In the display panel of the present application, the covering portion comprises a plurality of sub-covering portions at least covering one of the touch wirings.
[0019] In the display panel of the present application, a plurality of barrier sets are disposed in at least one of the barrier notches and are arranged along a second direction, and each of the barrier sets comprises a plurality of barrier bodies arranged along the first direction; [0020] wherein the barrier bodies in adjacent two of the barrier sets are staggered, and an included angle range between the first direction and the second direction is 0° to 90°.
[0021] In the display panel of the present application, along a direction of a top view of the display panel, a shape of each of the barrier bodies is one of a square, a rhombus, and a circle.
[0022] In the display panel of the present application, in adjacent two of the barrier sets, a central interval between adjacent two of the barrier bodies is 6 microns to 9 microns; [0023] wherein a length of a diagonal line of the barrier bodies is 8 microns to 12 microns.
[0024] In the display panel of the present application, along the second direction, a size of each of the barrier notches is equal, and an interval between adjacent two of the barrier notches is equal.
[0025] In the display panel of the present application, along the second direction, a size of each of the barrier notches is 40 microns, and an interval between adjacent two of the barrier notches is 40 microns.
[0026] In the display panel of the present application, along the second direction, a size of each of the barrier notches is greater than a size of the first aperture.
[0027] In the display panel of the present application, the first type recess comprises a first recess and a second recess, the second type recess comprises the two sub-recesses separated from each other, the second recess is located between the first recess and the sub-recesses, the sub-recesses are located on a side of the second recess away from the display region, and the first recess is located on a side of the second recess near the display region; [0028] wherein a boundary of the second refractive index layer is located between the first apertures and the first recess; [0029] or, a boundary of the second refractive index layer is located between the first recess and the second recess; [0030] or, a boundary of the second refractive index layer is located between the sub-recesses and the second recess.
[0031] In the display panel of the present application, along the first direction, a size of each of the sub-recesses is 1000 microns to 2000 microns.
[0032] In the display panel of the present application, the first recess or the second recess comprises a first barrier segment and a second barrier segment, the first barrier segment is

perpendicular to the second barrier segment, and an extension direction of the first barrier segment is parallel to the first direction; [0033] wherein along the first direction, an interval between the second barrier segment and the sub-recesses is 1000 microns to 2000 microns.
[0034] In the display panel of the present application, the first type recess is disposed between the display region and a bending region; [0035] wherein two ends of the first type recess extend to a boundary of the display panel and contacts the boundary of the display panel.
[0036] The present application also provides a mobile terminal, wherein the mobile terminal comprises a terminal main body and the above display panel, and the terminal main body and the display panel are assembled integrally.
Advantages
[0037] The present application disposes at least one sub-recess, not overlapping touch wirings, on a side away from the display region to reduce an overlapped region between barrier notches and the touch wirings and mitigate the technical issue of failure of the touch wirings due to over etching of the barrier notches.

Description

DESCRIPTION OF DRAWINGS
[0038] FIG. **1** is a first structural top view of a conventional display panel;
[0039] FIG. **2** is a second structural top view of the conventional display panel;
[0040] FIG. **3** is a third structural top view of the conventional display panel;
[0041] FIG. **4** is a first structural top view of the display panel of the present application;
[0042] FIG. **5** is a first structural cross-sectional view of the display panel of the present application;
[0043] FIG. **6** is a structural view of film layers of the display panel of the present application;
[0044] FIG. **7** is an enlarged view of barrier notches of the display panel of the present application;
[0045] FIG. **8** is a second structural cross-sectional view of the display panel of the present application;
[0046] FIG. **9** is a third structural cross-sectional view of the display panel of the present application;
[0047] FIG. **10** is a first enlarged view of a region AA in FIG. **4**; and
[0048] FIG. **11** is a second enlarged view of the region AA in FIG. **4**.
DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS
[0049] To make the objective, the technical solution, and the effect of the present application clearer and more explicit, the present application will be further described in detail below with reference to the accompanying drawings. It should be understood that the specific embodiments described here are only used to explain the present application instead of being used to limit the present application.
[0050] With reference to FIG. **1**, a square structure in FIG. **1** is a structure before a display panel is cut. An outer solid line of the square structure is a first cutting line GG. An inner broken line of the square structure in FIG. **1** is a second cutting line DD. The first cutting line GG is a cutting line formed by cutting a motherboard into the square structure. The second cutting line DD is a target shape of a display panel to be cut.
[0051] With reference to FIG. **2**, a micro array pattern in a MLP technology is usually required to be manufactured by an inkjet printing process. An irregular solid line in the square structure in FIG. **2** is an irregular boundary YM of an oil ink. With reference to FIG. **3**, for controlling a boundary of the oil ink, at present, usually a plurality of barrier notches **62** are defined in an outer periphery of display panel. However, because a structure of a lower boundary is different from structures of three other boundaries, a current barrier structure probably overlaps vertically disposed touch

wirings. Therefore, over etching occurs when the barrier notches **62** are etched such that the touch wirings are exposed to result in a of technical issue of touch failure. Thus, the present application sets forth a display panel to solve the above technical issue.

[0052] With reference to FIGS. **4** to **11**, the present application provides a display panel **100**. The display panel **100** comprises a display region **200** and a non-display region **300** located on at least one side of the display region **200**.

[0053] In the present embodiment, the display panel **100** can further comprise a substrate **10**, a light emitting layer **30**, a touch layer **50**, a first refractive index layer **60**, and a second refractive index layer **70**.

[0054] In the present embodiment, the light emitting layer **30** is disposed on a side of the substrate **10**. The light emitting layer **30** comprises a plurality of light emitting pixels disposed in the display region **200**. The first refractive index layer **60** is disposed on a side of the light emitting layer **30**. The first refractive index layer **60** comprises a plurality of first apertures distributed in the display region **200** and corresponding to the light emitting pixels **61**. The second refractive index layer **70** is disposed on a side of the first refractive index layer **60** away from the substrate **10** and is filled in the first apertures **61**. A refractive index of the second refractive index layer **70** is greater than a refractive index of the first refractive index layer **60**.

[0055] In the present embodiment, the touch layer **50** is disposed on a side of the light emitting layer **30** away from the substrate **10**. The touch layer **50** comprises touch electrodes disposed in the display region **200** and touch wirings **51** electrically connected to the touch electrodes. The touch wirings **51** extend from the display region **200** to the non-display region **300**.

[0056] In the present embodiment, the first refractive index layer **60** further comprises a plurality of barrier notches **62** distributed in the non-display region **300** and extending along a first direction X. The barrier notches **62** comprises a first type recess D**1** and a second type recess D**2**. The first type recess D**1** covers the touch wirings **51**. The second type recess D**2** comprises at least one sub-recess **623** not overlapping the touch wirings **51**.

[0057] The present application disposes at least one sub-recess **623** on a side of away from the display region **200** not overlapping the touch wirings **51** to reduce a contact area between the barrier notches **62** and the touch wirings **51**, which mitigates the technical issue of failure of the touch wirings **51** resulting from the over etched barrier notches **62**.

[0058] Now technology solutions of the present application are described with specific embodiments.

[0059] With reference to FIG. **4**, the display panel **100** can comprise a display region **200** and a non-display region **300** located on at least one side of the display region **200**. It should be explained that the non-display region **300** can be a lower frame region of the display panel **100**.

[0060] With reference to FIGS. **5** and **6**, the display panel **100** can comprise a driver circuit layer **20** disposed on the substrate **10**, a pixel definition layer **80** disposed on the driver circuit layer **20**, the light emitting layer **30** disposed in the same layer with the pixel definition layer **80**, an encapsulation layer **40** disposed on the pixel definition layer **80**, a touch layer **50** disposed on the encapsulation layer **40**, the first refractive index layer **60** disposed on the touch layer **50**, and the second refractive index layer **70** disposed on the first refractive index layer **60**.

[0061] In the present embodiment, a material of the substrate **10** can be a material such as glass, quartz, or polyimide.

[0062] In the present embodiment, with reference to FIG. **6**, the driver circuit layer **20** can comprise a plurality of thin film transistors **21**, the thin film transistors **21** can be etching barrier type, rear channel etching type, or can according to positions of a gate electrode and an active layer be divided into structures such as bottom gate thin film transistors, top gate thin film transistors, and have no specific limit therefor. For example, the thin film transistors **21** shown in FIG. **6** are top gate type thin film transistors, the thin film transistors **21** can comprise a light shielding layer **211** disposed on the substrate **10**, a buffer layer **212** disposed on the light shielding layer **211**, an active layer **213** disposed on the buffer layer **212**, a gate insulation layer **214** disposed on the active layer **213**, a gate electrode layer **215** disposed on the gate insulation layer **214**, an interlayer insulation layer **216** disposed on the gate electrode layer **215**, a source and drain electrode layer **217** disposed on the interlayer insulation layer **216**, and a planarization layer **218** disposed on the source and drain electrode layer **217**.

[0063] In the present embodiment, with reference to FIG. **5**, the display panel **100** can further comprise an anode layer **31** disposed on the planarization layer **218**, a light emitting layer **30** disposed on the anode layer **31**, and a cathode layer **32** disposed on the light emitting layer **30**. The anode layer **31** comprises a plurality of anodes **311**, the pixel definition layer **80** comprises a plurality of pixel apertures corresponding to the anodes **311**, and each of the pixel apertures exposes an upper surface of a corresponding one of the anodes **311**. The light emitting layer **30** can comprise a plurality of light emitting pixels corresponding to the anodes **311**.

[0064] In the present embodiment, with reference to FIG. **5**, the encapsulation layer **40** covers the pixel definition layer **80**, and continuously covers a plurality of pixel apertures and the light emitting pixels. The encapsulation layer **40** can at least comprise a first inorganic encapsulation layer, a first organic encapsulation layer, and a second inorganic encapsulation layer stacked on the pixel definition layer **80**.

[0065] In the present embodiment, with reference to FIG. **5**, the touch layer **50** can comprise a first touch metal layer **502** and a second touch metal layer **504** disposed on the encapsulation layer **40**, and an insulation layer disposed between the first touch metal layer **502** and the second touch metal layer **504**.

[0066] In the present embodiment, the touch layer **50** provided by the embodiment of the present application can be mutual capacitive or self-capacitive.

[0067] In the present embodiment, when the touch layer **50** is mutual capacitive, then the first touch metal layer **502** can comprise a plurality of first electrodes and a plurality of second electrodes. The first electrodes are connected through first connection bridges in the first touch metal layer **502**. The second electrodes are connected through second connection bridges extending through the insulation layer in the second touch metal layer **504**. When the touch layer **50** is self-capacitive, the first touch metal layer **502** can comprise a plurality of touch electrodes distributed in an array, and the second touch metal layer **504** can comprise a plurality of touch wirings **51**, and the touch wirings **51** are connected correspondingly to the touch electrodes.

[0068] In the present embodiment, when the touch layer **50** is self-capacitive, the touch layer **50** can only comprise one metal layer. Namely, the metal layer comprises a plurality of touch electrodes and a plurality of touch wirings **51** located among adjacent ones of the touch electrodes, and the touch wirings **51** are connected correspondingly to the touch electrodes. Furthermore, the embodiment of the present application are only examples as described above but it not limited thereto. Specific types structures of the touch layer **50** can be selected according to actual demands.

[0069] In the present embodiment, with reference to FIG. **5**, the first refractive index layer **60** can be disposed on the touch layer **50**, and the first refractive index layer **60** covers the display region **200** of the display panel **100** and extends to the non-display region **300**. The first refractive index layer **60** can comprise a plurality of first apertures **61** formed in the display region **200**, a position below each of the first apertures **61** corresponds to one of the pixel apertures, namely, the first apertures **61** corresponds to the light emitting pixels, and cross-section of the first apertures **61** can be inverted-trapezoidal.

[0070] In the present embodiment, the first refractive index layer **60** can further comprise at least one of the barrier notches **62** in the non-display region **300**. Each of the barrier notches **62** comprises the barrier bodies **63**, and a number of the barrier notches **62** can be one, two, three, or more. Each of the barrier notches **62** is configured to prevent the second refractive index layer **70** from spreading toward non-display region **300**.

[0071] In the present embodiment, with reference to FIG. **5**, the second refractive index layer **70** can be disposed on the first refractive index layer **60**, and the second refractive index layer **70** also covers display region **200** and extends to the non-display region **300**. The second refractive index layer **70** is filled in the first apertures **61** to form a plurality of micro lens units in the first apertures **61** to further perform a light convergence effect to the light emitting pixels corresponding to micro lens units and improve light emission effect of the corresponding light emitting pixels to further enhance a light emission rate of the display panel **100**.

[0072] In the present embodiment, the second refractive index layer **70** is also spread to the non-display region **300**, and is filled in at least some of the barrier bodies **63** in the barrier notches **62**.

[0073] In the present embodiment, the refractive index of the second refractive index layer **70** can be greater than the refractive index of the first refractive index layer **60**. Light of a large angle emitted from the light emitting pixels enters the first apertures **61** in the first refractive index layer **60** through the encapsulation layer **40**, and then enters an interface between the first refractive index layer **60** and the second refractive index layer **70**. Because the refractive index of the second refractive index layer **70** is greater than the refractive index of the first refractive index layer **60**, the light of a large angle entering the interface between the first refractive index layer **60** and the second refractive index layer **70** would undergo a total reflection to achieve a light convergence effect corresponding to the light emitting pixels, improve a light emission effect of corresponding light emitting pixels to further enhance a light emission rate of the display panel **100**.

[0074] In the present embodiment, one of the first apertures **61** in the first refractive index layer **60** forms a micro structure with a light converging effect.

[0075] In the present embodiment, a material of the first refractive index layer **60** and the second refractive index layer **70** can be a high light transmissive material, and a transmittance thereof is generally required to be greater than 90%. Namely, the transmittance of the first refractive index layer **60** and the second refractive index layer **70** is far greater than a transmittance of a general polarizer (42%). Disposing a material with a high transmittance for replacing the polarizer can effectively improve a light emission rate of the display panel **100**.

[0076] In the present embodiment, a refractive index of the first refractive index layer **60** can be 1.4 to 1.6, and a material of the first refractive index layer **60** can comprise a light transmissive organic material with a low refractive index. For example, a material of the first refractive index layer **60** can be acrylic acid resin, polyimide resin, polyamide resin and/or Alq3[Tris (8-hydroxyquinolinato) aluminum], etc.

[0077] In the present embodiment, a refractive index of the second refractive index layer **70** can be 1.61 to 1.8, and a material of the second refractive index layer **70** can comprise a light transmissive organic material including a high refractive index. For example, a material of the second refractive index layer **70** can be Poly (3,4-ethylenedioxythiophene) (PEDOT), 4,4′-Bis [N-(3-methylphenyl)-N-phenylamino]biphenyl (TPD), 4,4′,4″-Tris (N-3-methylphenyl-N-phenylamino) triphenylamine (m-MTDATA), 1,3,5-tris [N,N-bis(2-methylphenyl)-amino]-benzene (o-MTDAB), 1,3,5-tris [N,N-bis(3-methylphenyl)-amino]-benzene (m-MTDAB), 1,3,5-tris [N,N-bis(4-methylphenyl)-amino]-benzene (p-MTDAB), 4,4′-bis [N,N-bis(3-methylphenyl)-amino]-diphenylmethane (BPPM), 4,4′-N,N′-dicarbazol-biphenyl (CBP), 4′4′,4″-tris(carbazol-9-yl)-triphenylamine (TCTA), 2,2′, 2″-(1,3,5-phenylene) tris-[1-phenyl-1H-benzimidazole] (TPBI), and 3-(4-biphenyl)-4-phenyl-5-tert-butylphenyl-1,2,4-triazole (TAZ).

[0078] In the display panel **100** of the present application, because a structure of the lower boundary is different from structures of other three boundaries, the current barrier structure cannot fulfill control of an oil ink on the cutoff boundary of the lower boundary. Therefore, the present application patterns the barrier notches **62** near lower frame region, namely, a regular hollowed and continuous recess is replaced with the barrier bodies **63** of the present application.

[0079] With reference to FIG. **4**, the barrier notches **62** in a region AA of FIG. **4** requires a new design. With reference to FIG. **7**, FIG. **7** is a schematic view of one barrier notch **62** in a plurality of barrier notches **62**. A plurality of barrier sets **65** along a second direction Y are disposed in the barrier notches **62**. Each of the barrier sets **65** comprises the barrier bodies **63** disposed along the first direction X. The barrier bodies **63** in adjacent two of the barrier sets **65** are staggered, and an included angle range between the first direction X and the second direction Y is 0° to 90°.

[0080] In the present embodiment, the first direction X can be the extension direction of the barrier notches **62** and can be parallel to a lower frame of the display panel **100**. The second direction Y can be a direction from the display region **200** to the non-display region **300**, namely, the first direction X can be perpendicular to the second direction Y.

[0081] In the present embodiment, the barrier bodies **63** are arranged along the first direction X and the second direction Y, and the barrier bodies **63** of adjacent two rows are staggered. An interval between adjacent two of the barrier bodies **63** in one barrier set **65** is less than a size of the barrier bodies **63** along the first direction X.

[0082] In the present embodiment, the barrier bodies **63** can be protrusion structures disposed in the barrier notches **62**. For example, when the barrier notches **62** are etched and hollowed, some of protrusion structures are retained to increase a contact area between the second refractive index material and the barrier notches **62** to further improve a surface tension of the second refractive index material. Furthermore, the barrier bodies **63** can also be aperture structures defined in the barrier notches **62**. for example, a film layer structure that a recess depth formed by the barrier notches **62** not etched to the touch layer **50** such that holes are defined one the basis of the film layer to increase a contact area between the second refractive index material and the barrier notches **62** to further improve a surface tension of the second refractive index material.

[0083] In the present embodiment, when the barrier bodies **63** are protrusion structures, a depth of each of the barrier bodies **63** is the same as a thickness of the first refractive index layer **60**. When the barrier bodies **63** is an aperture structure, a sum of the depth of each of the barrier bodies **63** and the depth of the barrier notch is the same as the thickness of the first refractive index layer **60**.

[0084] The present embodiment replaces a continuous recess hollowed in the first refractive index layer **60** with a plurality of non-continuous barrier bodies **63** to increase the contact area between the second refractive index layer **70** and the first refractive index layer **60** to further increase a surface tension of the second refractive index layer **70** and reduce an expansion force of the second refractive index layer **70** toward the non-display region **300**, which effectively controls the boundary of the second refractive index layer **70** and saves a space of the non-display region **300** to achieve narrow frame display.

[0085] In the display panel **100** of the present application, along a direction of a top view of the display panel **100**, a shape of each of the barrier bodies **63** is of a square, a rhombus, and a circle.

[0086] With reference to FIG. **7**, a shape of the barrier bodies **63** can be a square rotating by 45°.

[0087] In the present embodiment, when the second refractive index layer **70** spreads from the display region **200** of the display panel **100** to the non-display region **300**, the second refractive index layer **70** would pass through at least some of the barrier bodies **63**. Because the barrier bodies **63** is a square rotating by 45°, an included angle is defined between a flowing direction of the second refractive index layer **70** and the boundary of the barrier bodies **63**, and more resistance points exist, which can increase a resistance force of the barrier bodies **63** to the second refractive index layer **70**.

[0088] In the present embodiment, although a regular square can also effectively control the cutoff boundary of the second refractive index layer **70**, compared to the solution in FIG. **7**, the regular square has two boundaries parallel to the flowing direction of the second refractive index layer **70**, and its resistance force to the second refractive index layer **70** is less than that of the solution in FIG. **7**.

[0089] In the display panel **100** of the present application, considering a size of the barrier notches **62** and a number of the barrier bodies **63**, in adjacent two of the barrier sets **65**, an interval of a central point of adjacent two of the barrier bodies **63** can be 6 microns to 9 microns, each of the barrier bodies **63** □ a length of a diagonal line is 8 microns to 12 microns.

[0090] With reference to FIG. **7**, a square rotating by 45° is taken as an example, a size of two diagonal lines of the barrier bodies **63** can be 10 microns, a size of any edge of the barrier bodies **63** is 7.07 microns.
[0091] In the present embodiment, with reference to FIG. **7**, in the different barrier sets **65**, an interval of adjacent two of the barrier bodies **63** can be half a side length of the barrier bodies **63**, i.e., 3.54 microns. Second, in adjacent two of the barrier sets **65**, an interval of a central point of adjacent two of the barrier bodies **63** along the second direction Y can be 7.5 microns.
[0092] In the present embodiment, a size of the barrier notches **62** along the second direction Y can be 40 microns.
[0093] In the display panel **100** of the present application, a structure shown in FIG. **4** is taken as an example, the first refractive index layer **60** can comprise two the first type recesses D**1** and one second type recess D**2**. The two first type recesses D**1** can be a first recess **621** and a second recess **622**. The second type recess D**2** can comprise two sub-recesses **623**. The second recess **622** is located between the first recess **621** and the sub-recesses **623**, the sub-recesses **623** are located on a side of the second recess **622** away from the display region **200**, and the first recess **621** is located on a side of the second recess **622** near the display region **200**.
[0094] With reference to FIG. **8**, the cutoff boundary of the second refractive index layer **70** is located between the first apertures **61** and the first recess **621**. When the second refractive index **70** extends toward the first recess **621**, probably due to a sufficient surface tension of the second refractive index **70** itself, the second refractive index **70** does not cover the first recess **621** such that the cutoff boundary of the second refractive index **70** is located between the first recess **621** and the display region **200**.
[0095] With reference to FIG. **9**, the cutoff boundary of the second refractive index **70** can be located between the second recess **622** and the first recess **621**. When the second refractive index **70** extends toward the second recess **622**, probably due to a barrier function of the first recess **621** and a sufficient surface tension of the second refractive index **70** itself, the second refractive index **70** does not cover the second recess **622** such that the cutoff boundary of the second refractive index **70** is located between the first recess **621** and the second recess **622**.
[0096] With reference to FIG. **5**, the cutoff boundary of the second refractive index layer **70** can be located between the sub-recess **623** and the second recess **622**. When the second refractive index layer **70** extends toward the sub-recess **623**, probably due to a barrier function of the second recess **622** and a sufficient surface tension of the second refractive index layer **70** itself, the second refractive index layer **70** does not cover the second recess **622** such that the cutoff boundary of the second refractive index layer **70** is located between the sub-recess **623** and the second recess **622**.
[0097] Furthermore, the second refractive index layer **70** can also cover the sub-recess **623** and extend along a direction away from the sub-recess **623**.
[0098] In the present embodiment, to prevent the second refractive index layer **70** from further extending along a direction away from the display region **200** in the display panel **100** after crossing the second recess **622**, and failing to effectively controlling the cutoff boundary of the second refractive index layer **70**, the sub-recess **623** is also disposed on a side of the second recess **622** away from the display region **200**. The sub-recess **623** can comprise a plurality of non-continuous the barrier bodies **63**.
[0099] In the present embodiment, the barrier bodies **63** in the first recess **621**, the second recess **622**, and the sub-recess **623** cooperatively function to further increase a surface tension of the second refractive index layer **70** and reduce an expansion force of the second refractive index layer **70** toward the non-display region **300** such that the cutoff boundary of the second refractive index is effectively controlled.
[0100] In the present embodiment, along the first direction X, a size C of each of the sub-recess **623** can be 1000 microns to 2000 microns.
[0101] In the present embodiment, with reference to FIG. **10**, along the first direction X, the size C

of each of the sub-recess **623** can be 2000 microns.

[0102] In the display panel **100** of the present application, with reference to FIG. **10**, the first recess **621** or the second recess **622** comprises a first barrier segment **621***a* and a second barrier segment **621***b*. The first barrier segment **621***a* is perpendicular to the second barrier segment **621***b*. An extension direction of the first barrier segment **621***a* is parallel to the first direction X. The structure in FIG. **10** marks the second recess **622** as an example.

[0103] In the present embodiment, along the first direction X, a first interval A between the second barrier segment **621***b* and the sub-recess **623** is 1000 microns to 2000 microns.

[0104] In the present embodiment, with reference to FIG. **10**, the first interval A can be 2000 microns.

[0105] With reference to FIG. **10**, because the broken line in FIG. **10** is a second cutting line DD, namely, a region enclosed by the second cutting line DD is a final shape of the display panel **100**, a structure outside the broken line will no longer exist in the final product. However, a second interval B between the first barrier segment **621***a* in the first recess **621** and a corner region is generally set as 300 microns to 500 microns.

[0106] In the present embodiment, with reference to FIG. **10**, the second interval B can be 500 microns.

[0107] In the display panel **100** of the present application, with reference to FIGS. **5** and **11**, the touch layer **50** can be disposed between the encapsulation layer **40** and the first refractive index layer **60**. The touch layer **50** comprises a plurality of touch wirings **51** extending from the display region **200** to the non-display region **300**. The touch wirings **51** comprises a plurality of overlapping portions overlapping at least one of the barrier notches **62**.

[0108] In the present embodiment, with reference to FIG. **5**, the touch layer **50** can comprise a first insulation layer **501** disposed on the encapsulation layer **40**, a first touch metal layer **502** disposed on the first insulation layer **501**, a second insulation layer **503** disposed on the first touch metal layer **502**, and a second touch metal layer **504** disposed on the second insulation layer **503**. The second insulation layer **503** covers the first touch metal layer **502**. The first refractive index layer **60** covers the second touch metal layer **504**. In FIG. **11**, when the touch wirings **51** extend from the display region **200** to the non-display region **300**, the touch wirings **51** would cross the first recess **621** or/and the second recess **622**. Therefore, a covering portion **64** is filled in an overlapped position between the touch wirings **51** and the first type recess D**1**.

[0109] In the present embodiment, the covering portion **64** can be a material of the first refractive index layer **60** not etched.

[0110] With reference to FIG. **11**, the covering portion **64** comprises a plurality of sub-covering portions **641** at least covering one of the touch wirings **51**. In the present embodiment, one sub-covering portion **641** covers one touch wiring **51**, and the barrier bodies **63** can be disposed between adjacent two of the sub-covering portions **641**.

[0111] With reference to FIGS. **5** and **11**, because the first refractive index layer **60** covers the second touch metal layer **504**, when the touch wirings **51** is the second touch metal layer **504**, the barrier bodies **63** would interfere with the touch wirings **51**. Therefore, to prevent interference of both, the present application disposes sub-covering portions **641** corresponding to the touch wirings **51** in the first recess **621** or/and the second recess **622**. The sub-covering portions **641** on the first recess **621** or/and the second recess **622** have no barrier bodies **63**, which prevent interference of the barrier bodies **63** with touch wirings **51**.

[0112] In the present embodiment, when the touch wirings **51** is the first touch metal layer **502**, because a thickness of the second insulation layer **503** is thinner, a risk of over etching easily occurs when the first refractive index layer **60** is etched to form the barrier bodies **63**.

[0113] The present embodiment employs a design of omitting barrier bodies **63** in regions overlapping the touch wirings **51** to prevent interference with the touch wirings **51** during etching the barrier bodies **63**, which avoids the technical issue of over etching the touch wirings **51**.

[0114] In the present embodiment, the driver circuit layer **20** can further comprise a plurality of driver circuits electrically connected to corresponding ones of the light emitting pixels and driver wirings electrically connected to the driver circuits. The driver wirings extend from the display region **200** to the non-display region **300**. The non-display region **300** comprises a bending region **400** and a wiring exchange region **500** disposed between the bending region **400** and the display region **200**. The touch wirings **51** are electrically connected to the driver wirings through via holes **52** in the wiring exchange region **500**, one wiring exchange region **500** is disposed between adjacent two of the sub-recesses **623**.
[0115] With reference to FIG. **11**, the touch wirings **51** extending from the display region **200** to the non-display region **300**. To reduce touch driver chips that are set, usually the touch wirings **51** in the touch layer **50** are exchanged into an array layer to use a chip to perform display driving and touch driving simultaneously. At the same time, along a direction of a top view of the display panel **100**, the touch wirings **51** is disposed between adjacent two of the sub-recesses **623**. Because the sub-recesses **623** are disposed separately, it is only required that the interval between adjacent two of the sub-recesses **623** is guaranteed to allow the touch wirings **51** to pass through the interval.
[0116] In the display panel **100** of the present application, with reference to FIG. **4**, in the barrier notches **62**, along the second direction Y, a size of each of the barrier notches **62** can be equal, an interval between adjacent two of the barrier notches **62** can be equal.
[0117] In the display panel **100** of the present application, in the barrier notches **62**, along the second direction Y, the size of each of the barrier notches **62** can be 40 microns, the interval between adjacent two of the barrier notches **62** can be 40 microns.
[0118] For example, the first recess **621**, a size of the second recess **622** and a size of the sub-recess **623** along the second direction Y can be equal, the size can be 40 microns. Adjacent intervals among the first recess **621**, the second recess **622**, and the sub-recess **623** along the second direction Y can be equal, the interval can be 40 microns.
[0119] In the display panel **100** of the present application, along the second direction Y, a size of each of the barrier notches **62** is greater than a size of the first aperture **61**. The present embodiment sets the sizes of the first recess **621**, the second recess **622** and the sub-recess **623** along the second direction Y greater to be able to more effectively prevent ink in th process from flowing.
[0120] In the display panel **100** of the present application, with reference to FIG. **10**, the non-display region **300** comprises a bending region **400** configured for bending. The barrier notches **62** is disposed between the display region **200** and the bending region **400**. Because the bending region **400** of the display panel **100** corresponds to a lower frame of a final product, a position of the bending region **400** is equivalent to a position of the lower frame of the final product. The present embodiment disposes the barrier notches **62** between the display region **200** and the bending region **400**, limits the cutoff boundary of the second refractive index between the display region **200** and the bending region **400** through the barrier notches **62** to achieve a narrow frame design.
[0121] In the present embodiment, two ends of the first type recess D**1** extend to the boundary of the display panel **100** and contact the boundary of the display panel **100**. With reference to FIG. **10**, two ends of the first recess **621** and two ends of the second recess **622** can extend toward the boundary of the display panel **100** and contact the boundary of the display panel **100**. Furthermore, a certain interval exists between the sub-recess **623** and the display panel **100**.
[0122] The present application also sets forth a mobile terminal comprising a terminal main body and the above display panel. The terminal main body and the display panel are assembled integrally. The terminal main body can be bonded to a device of the display panel such as a circuit board and covers a cover lid of the display panel. The mobile terminal can comprise an electronic apparatus such as cell phone, television, notebook, etc.
[0123] It can be understood that for a person of ordinary skill in the art, equivalent replacements or changes can be made according to the technical solution of the present application and its inventive

concept, and all these changes or replacements should belong to the scope of protection of the appended claims of the present application.

Claims

1. A display panel, wherein the display panel comprises a display region and a non-display region located on at least one side of the display region; the display panel further comprises: a substrate; a light emitting layer disposed on a side of the substrate, wherein the light emitting layer comprises a plurality of light emitting pixels disposed in the display region; a touch layer disposed on a side of the light emitting layer away from the substrate, wherein the touch layer comprises touch electrodes disposed in the display region and touch wirings electrically connected to the touch electrodes, and the touch wirings extend from the display region to the non-display region; a first refractive index layer disposed on a side of the touch layer away from the substrate, wherein the first refractive index layer comprises a plurality of first apertures distributed in the display region and corresponding to the light emitting pixels; a second refractive index layer disposed on a side of the first refractive index layer away from the substrate, filled in the first apertures, and a refractive index of the second refractive index layer is greater than a refractive index of the first refractive index layer; wherein the first refractive index layer further comprises a plurality of barrier notches distributed in the non-display region and extending along a first direction, the barrier notches comprise a first type recess and a second type recess, the first type recess is disposed to cover the touch wirings, and the second type recess comprises at least two sub-recesses spaced from each other at intervals.
2. The display panel according to claim 1, wherein each of the at least two sub-recesses does not overlap the touch wirings; and the display panel further comprises a driver circuit layer disposed between the substrate and the light emitting layer, wherein the driver circuit layer comprises a plurality of driver circuits electrically connected to corresponding ones of the light emitting pixels and driver wirings electrically connected to the driver circuits, and the driver wirings extend from the display region to the non-display region; wherein the non-display region comprises a bending region and a wiring exchange region disposed between the bending region and the display region, wherein the touch wirings are electrically connected to the driver wirings through via holes in the wiring exchange region; wherein a wiring exchange region is disposed between adjacent two of the sub-recesses.
3. The display panel according to claim 1, wherein each of the at least two sub-recesses does not overlap the touch wirings; and an overlapped position between the first type recess and the touch wirings are filled with a covering portion.
4. The display panel according to claim 3, wherein the covering portion comprises a plurality of sub-covering portions at least covering one of the touch wirings.
5. The display panel according to claim 1, wherein a plurality of barrier sets are disposed in at least one of the barrier notches and are arranged along a second direction, and each of the barrier sets comprises a plurality of barrier bodies arranged along the first direction; wherein the barrier bodies in adjacent two of the barrier sets are staggered, and an included angle range between the first direction and the second direction is 0° to 90°.
6. The display panel according to claim 5, wherein along a direction of a top view of the display panel, a shape of each of the barrier bodies is one of a square, a rhombus, and a circle.
7. The display panel according to claim 5, wherein in adjacent two of the barrier sets, a central interval between adjacent two of the barrier bodies is 6 microns to 9 microns.
8. The display panel according to claim 7, wherein a length of a diagonal line of the barrier bodies is 8 microns to 12 microns.
9. The display panel according to claim 8, wherein along the second direction, a size of each of the barrier notches is equal, and an interval between adjacent two of the barrier notches is equal.

10. The display panel according to claim 8, wherein along the second direction, a size of each of the barrier notches is 40 microns, and an interval between adjacent two of the barrier notches is 40 microns.
11. The display panel according to claim 8, wherein along the second direction, a size of each of the barrier notches is greater than a size of the first aperture.
12. The display panel according to claim 8, wherein the first type recess comprises a first recess and a second recess, the second type recess comprises the two sub-recesses spaced from each other, the second recess is located between the first recess and the sub-recesses, the sub-recesses are located on a side of the second recess away from the display region, and the first recess is located on a side of the second recess near the display region.
13. The display panel according to claim 12, wherein a boundary of the second refractive index layer is located between the first apertures and the first recess.
14. The display panel according to claim 12, wherein a boundary of the second refractive index layer is located between the first recess and the second recess.
15. The display panel according to claim 12, wherein a boundary of the second refractive index layer is located between the sub-recesses and the second recess.
16. The display panel according to claim 12, wherein the first recess or the second recess comprises a first barrier segment and a second barrier segment, the first barrier segment is perpendicular to the second barrier segment, and an extension direction of the first barrier segment is parallel to the first direction.
17. The display panel according to claim 12, wherein the first type recess is disposed between the display region and a bending region; wherein two ends of the first type recess extend to a boundary of the display panel and contacts the boundary of the display panel.
18. A display panel, wherein the display panel comprises a display region and a non-display region located on at least one side of the display region; the display panel further comprises: a substrate; a light emitting layer disposed on a side of the substrate, wherein the light emitting layer comprises a plurality of light emitting pixels disposed in the display region; and a first refractive index layer disposed on a side of the touch layer away from the substrate, wherein the first refractive index layer comprises a plurality of first apertures distributed in the display region and corresponding to the light emitting pixels; a second refractive index layer disposed on a side of the first refractive index layer away from the substrate, filled in the first apertures, and a refractive index of the second refractive index layer is greater than a refractive index of the first refractive index layer; wherein the first refractive index layer further comprises a plurality of barrier notches distributed in the non-display region and extending along a first direction, the barrier notches comprise a first type recess and a second type recess, the first type recess extends continuously along the first direction, and the second type recess comprises at least two sub-recesses arranged along the first direction and spaced from each other at intervals.
19. The display panel according to claim 18, wherein the display panel further comprises a touch layer disposed on a side of the light emitting layer away from the substrate, wherein the touch layer comprises touch electrodes disposed in the display region and touch wirings electrically connected to the touch electrodes, and the touch wirings extend from the display region to the non-display region; and the first type recess covers the touch wirings, and the second type recess comprises at least one of the sub-recesses not overlapping the touch wirings.
20. A mobile terminal, wherein the mobile terminal comprises a terminal main body and the display panel according to claim 1, and the terminal main body and the display panel are assembled integrally.